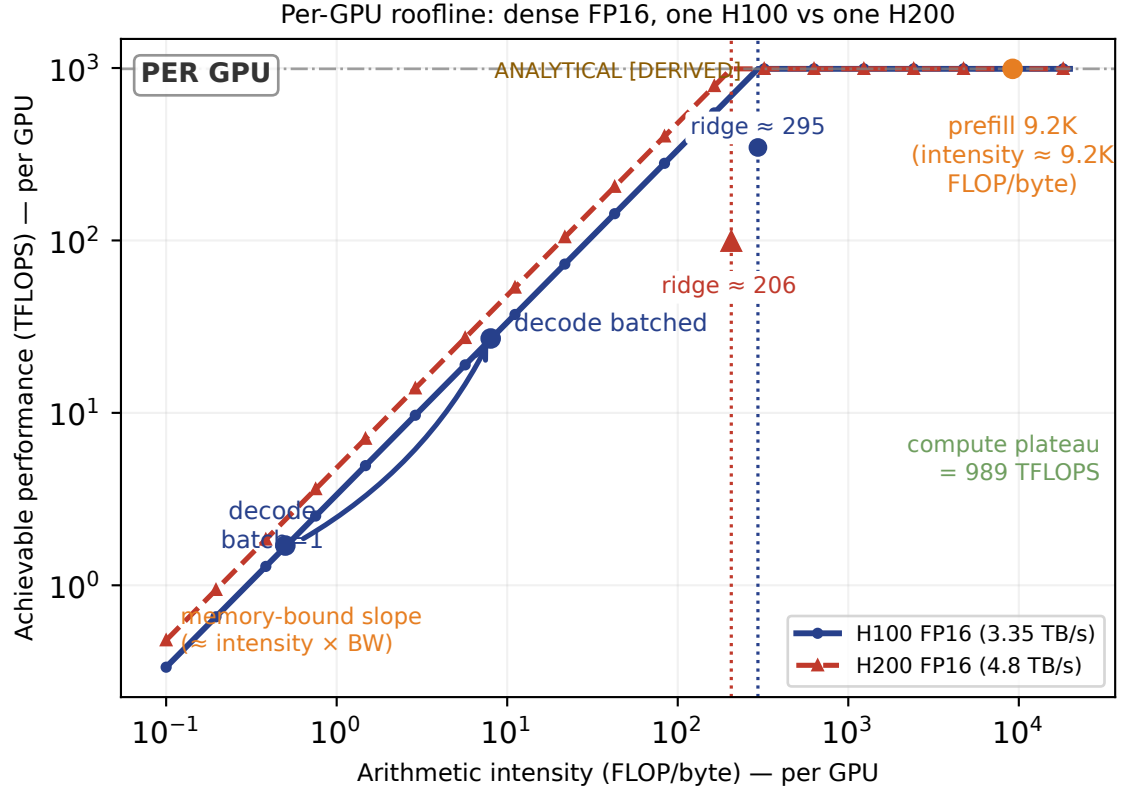
Per-GPU roofline: dense FP16, one H100 vs one H200
PER GPU
ANALYTICAL [DERIVED]
ridge ≈ 295
prefill 9.2K
(intensity ≈ 9.2K
FLOP/byte)
ridge ≈ 206
decode batched
compute plateau
= 989 TFLOPS
decode
batch=1
memory-bound slope
(≈ intensity × BW)
H100 FP16 (3.35 TB/s)
H200 FP16 (4.8 TB/s)
Achievable performance (TFLOPS) — per GPU
Arithmetic intensity (FLOP/byte) — per GPU
10^3
10^2
10^1
10^0
10^{-1}
10^0
10^1
10^2
10^3
10^4